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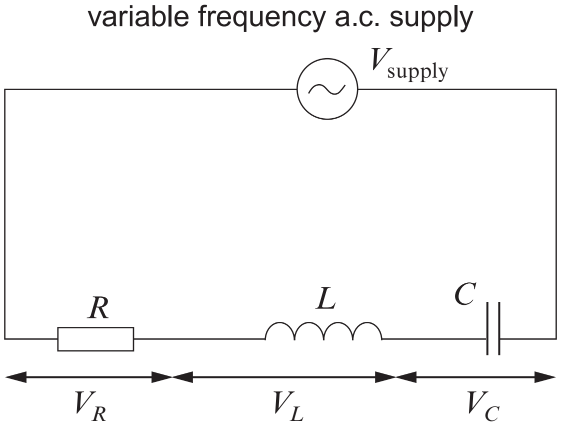
Option A – **Alternating Currents**

6. *(a)* The following RCL circuit is constructed.



For this RCL circuit, describe the relationships between the rms pds V_{supply}, V_R, V_L and V_C:

(i) when the circuit is not in resonance; [3]
Space for diagram.

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(ii) when the circuit is in resonance. [2]

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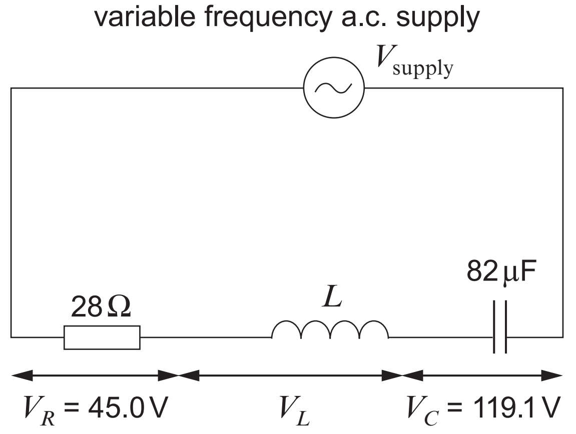
(b) The following RCL circuit is **at resonance**. The values of R and C are provided along with their rms pd values.



Calculate:

(i) the Q factor of the circuit; [1]

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(ii) the rms current; [1]

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(iii) the frequency of the power supply; [2]

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(iv) the inductance of the inductor. [2]

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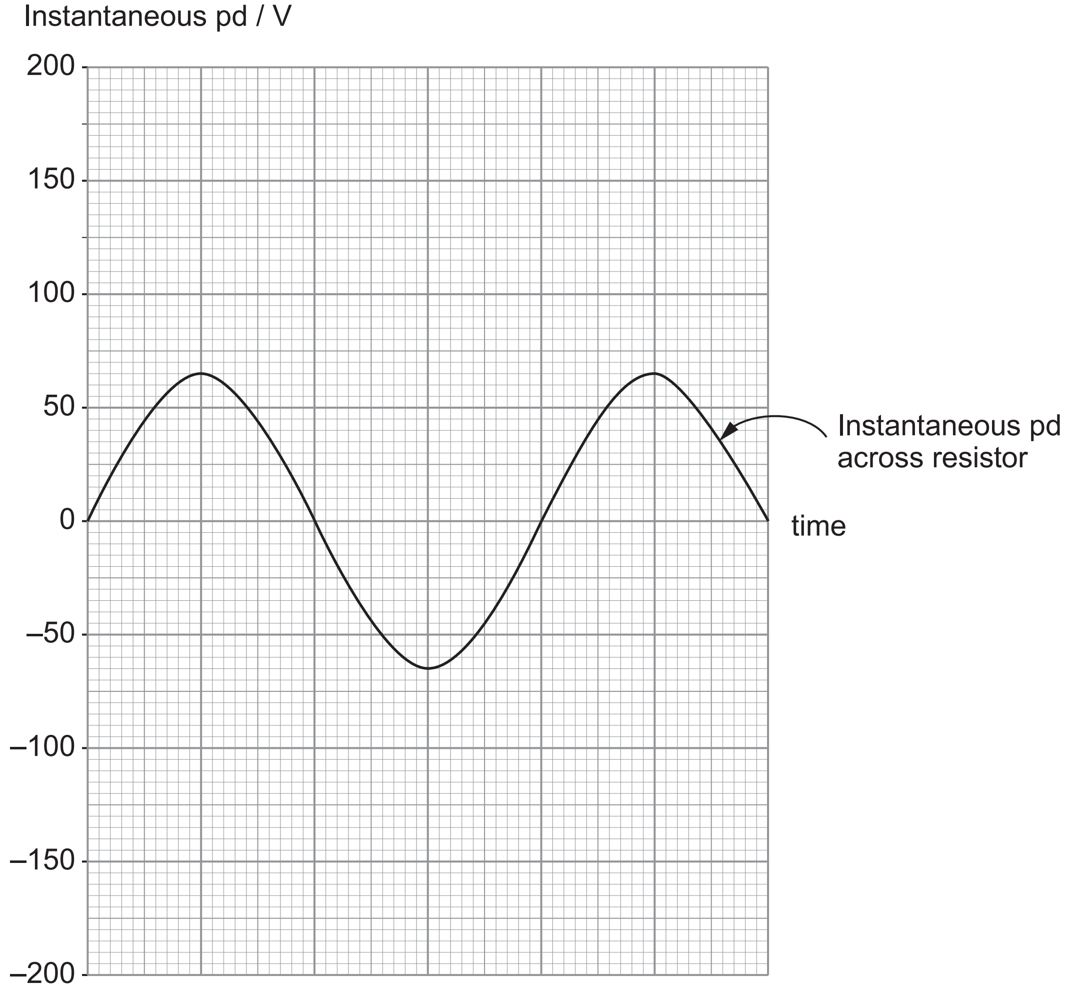
(v) Sketch a graph of the instantaneous pd across the capacitor on the grid provided. (The instantaneous pd across the resistor is shown.) [2]



(vi) Without further calculation, explain why the current decreases when the frequency of the power supply is increased. [2]

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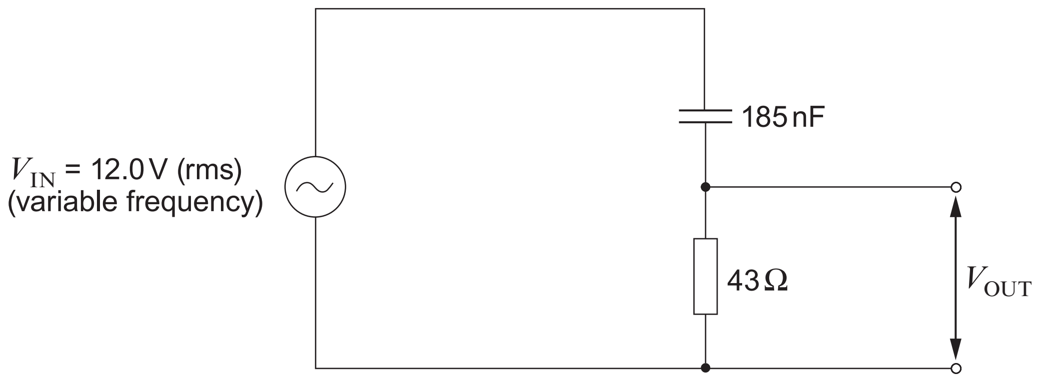
(c) Sam claims that the output pd (V_{OUT}) in the following circuit is greater than 6.0 V when the frequency is greater than 20 kHz, but less than 6.0 V when below 20 kHz. Investigate whether or not Sam is correct. [5]



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